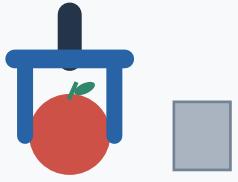
01 HSR: task state

Approach → grasp recognized



HSR estimates task phase

and whether the object is held.

grasp / held signal

02 LTE: link-specific target exemption

Before attachment • two views of the same scene

AUTHORIZED LINKS

Fingers / palm

target-free field

OTHER LINKS

Wrist / arm

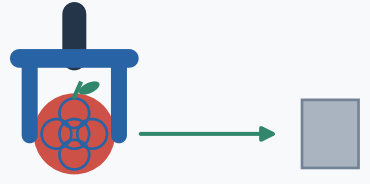
full-scene field

Target contact allowed for designated links only

Other obstacles remain in both fields

03 After grasp

Object joins robot collision geometry



attached spheres environment

Held volume removed from field;

moves with hand and is checked

against the environment.

Conceptual schematic • Dashed red: target excluded from queried field • Gray: other obstacles